

Trading Advisor

System Specifications & Strategy Guide

A Quantitative Multi-Regime Framework for NSE Instruments

Modes: ETF Swing • Stock Delivery • Intraday ORB

Generated: July 2026

1. Executive System Overview

The Trading Advisor system is a fully automated, quantitative scanning and position management framework designed for National Stock Exchange (NSE) instruments. The platform operates across three distinct trading horizons (Modes) to optimize capital allocation and risk adjusted returns:

- **Mode A: ETF / Mutual Fund Swing Trading** — Focused on mean-reversion anomalies in index and sector-level ETFs. Run frequency: Daily.
- **Mode B: Stock Delivery Positional Trading** — Focuses on trend pullback entries in high-quality large and mid-cap stocks. Run frequency: Every 2 hours.
- **Mode C: Intraday Opening Range Breakout (ORB)** — Capitalizes on short-term breakout momentum during session open. Run frequency: Every 30 minutes.

Market Horizons Comparison

Parameter	Mode A: ETF Swing	Mode B: Stock Delivery	Mode C: Intraday ORB
Time Horizon	3 to 10 Days	10 to 20 Days	Same-day (No Overnight)
Primary Strategy	Mean Reversion	Momentum Pullback	Opening Range Breakout
Allocation Corpus	■10,000	■10,000	■5,000
Max Positions	2 concurrent	2 concurrent	2 concurrent
Data Interval	Daily (1D)	Daily (1D)	5-minute (5m)

CRITICAL RISK LIMITATION

The system executes hard-coded risk management parameters. No single trade in Mode A or B is sized manually; instead, position sizing is dynamically computed using the Average True Range (ATR) over 14 periods to restrict exposure to a maximum of 5% of the respective mode's total capital.

2. Mode A: ETF / Mutual Fund Swing Trading

Mode A focuses on index and sectoral exchange-traded funds listed on the NSE. ETFs provide intrinsic diversification, reducing the threat of sudden corporate governance failures or company-specific earnings gaps.

Watchlist and Market Segment

The watchlist consists of liquid asset classes including Indian Sectoral/Thematic ETFs, global technology indices, and commodities:

- **Sectoral:** BANKBEES.NS, PSUBNKBEEES.NS, PHARMABEES.NS, ITETF.NS, AUTOBEES.NS, JUNIORBEES.NS
- **Commodities:** OILETF.NS (tracks Crude Oil)
- **International:** N100.NS (Nasdaq 100), MAFANG.NS (FANG+ Index), HNGSNGBEES.NS (Hang Seng)
- **High Beta Swing Stocks:** IRFC.NS, RVNL.NS, IREDA.NS, TATAPOWER.NS, SUZLON.NS

Signal Calculus & Scoring

The score is derived out of 100 points, calculated using the formula:

$$\text{ETF Score} = (0.25 * \text{RSI}) + (0.25 * \text{EMA_Cross}) + (0.15 * \text{Bollinger}) + (0.15 * \text{Volume}) + (0.10 * \text{News}) + (0.10 * \text{Macro})$$

- **RSI (25%):** Evaluates oversold states. RSI < 20 yields a signal score of +1.0. RSI > 70 yields -1.0.
- **EMA Cross (25%):** Detects short-term momentum shifts. Fresh bullish crossing of 9 EMA over 21 EMA gives +0.90.
- **Bollinger Bands (15%):** Measures price location inside volatility bands. Price < lower band gives +1.00 (oversold bounce candidate).
- **Volume Z-Score (15%):** Z-score of today's volume against 20-day average. Spike > 3.0 gives +1.00.

Rules for Entry and Exit

Rule Type	Condition	Threshold Value
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Entry	RSI Oversold Filter	RSI < 38
Entry	Volume Confirmation	Z-Score ≥ 1.5
Entry	Minimum Composite Score	$\geq 40 / 100$
Exit	Take Profit Target	+4.0% from entry
Exit	Stop Loss Limit	Entry - (1.5 * ATR14)
Exit	Time Out Exit	10 Trading Days max hold

3. Mode B: Stock Delivery Positional Trading

Mode B is configured to find momentum pullbacks in high-conviction, large and mid-cap stocks. Unlike Mode A, this mode does not buy oversold instruments (which are often "falling knives" in individual equities). Instead, it enters strong, structured uptrends on brief technical corrections.

Watchlist and Sector Mapping

The watchlist consists of high-beta, institutional-grade stocks. Every stock is mapped to a sectoral index to check relative strength:

- **Large Caps:** ICICIBANK.NS, HDFCBANK.NS, RELIANCE.NS
- **Quality Midcaps:** CDSL.NS, POLYCAB.NS, BAJFINANCE.NS, TITAN.NS, ANGELONE.NS, PIDILITIND.NS, KEI.NS, HAVELLS.NS

Relative Strength & Sector Gating

A core risk filter in Mode B is sector relative strength. If a stock belongs to the banking sector, it is mapped to ^NSEBANK. If that sector index is underperforming the Nifty 50 Index (^NSEI) by more than **3.0%**, the system suspends all entries in that stock, preventing capital exposure to structurally weak sectors.

Scoring Calculus

Calculated using the formula:

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Stock Score = (0.20 * RSI) + (0.30 * EMA_Cross) + (0.25 * Pullback) + (0.15 * Volume) + (0.05 * News) + (0.05 * Macro)
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- **RSI Pullback (20%):** Target range is RSI 42 to 55 (score +1.0). RSI < 38 is penalized as trend failure risk (score -0.3).
- **EMA Alignment (30%):** Bullish alignment (EMA9 > EMA21 > EMA50) yields +1.0.
- **EMA21 Pullback Proximity (25%):** Measures distance of price to 21-day EMA. Distance within -2% to +1% of EMA21 yields +1.0.
- **Volume Calm (15%):** Low volume on a dip indicates healthy consolidation. Volume Z-score < 0.5 yields +0.8. High volume dips are penalized.

Rules for Entry and Exit

Rule Type	Condition	Threshold Value
Entry	RSI Pullback Range	RSI between 42 and 55
Entry	EMA Proximity	Within 4.0% of daily EMA21
Entry	Nifty Trend Filter	Nifty 50 above its 20-EMA
Entry	Minimum Composite Score	$\geq 50 / 100$
Exit	Take Profit Target	+10.0% from entry (5% partial profit booking)
Exit	Stop Loss Limit	Entry - $(2.5 * ATR_{14})$

4. Mode C: Intraday Opening Range Breakout (ORB)

Mode C is designed to catch rapid intraday breakouts in high-liquidity stocks. All positions are entered, managed, and closed within a single session. Overnight exposure is strictly prohibited to avoid gap risk.

Intraday Watchlist

We restrict scanning to 12 highly liquid, institutional favorites: RELIANCE.NS, ICICIBANK.NS, HDFCBANK.NS, INFY.NS, TCS.NS, SBIN.NS, TATAMOTORS.NS, BAJFINANCE.NS, AXISBANK.NS, ADANIENT.NS, WIPRO.NS, POWERGRID.NS.

Opening Range Definition & Execution

The opening range is defined by the high and low of the first ****15 minutes**** of the session (9:15 AM to 9:30 AM IST).

- **ORB High:** The maximum high achieved in 9:15–9:30 IST.
- **ORB Low:** The minimum low achieved in 9:15–9:30 IST.
- **Trigger Level:** A 0.1% breakout buffer is applied above the ORB High to avoid whipsaws.

Intraday Scoring Calculus

Calculated using the formula:

$$\text{Intraday Score} = (0.40 * \text{ORB_Status}) + (0.25 * \text{Volume}) + (0.20 * \text{VWAP}) + (0.15 * \text{RSI})$$

- **ORB Status (40%):** Price sustaining above the breakout level yields +0.8 to +1.0.
- **Volume Multiplier (25%):** Volume of the breakout candle divided by the session's average 5-minute candle volume. Must be $\geq 1.5\times$ (score +0.5). Spike $\geq 3\times$ yields +1.0.
- **VWAP Proximity (20%):** Price must be above the Volume Weighted Average Price (VWAP) for long setups (score +0.35 to +1.0).
- **RSI Momentum (15%):** 5-minute RSI must be between 60 and 70 (+0.9).

Risk and Execution Parameters

Execution Metric	Value	Notes
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Intraday Corpus	■5,000	Separated from swing corpus
Max Risk Per Trade	2.0% (■100)	Restricts risk of ruin
Profit Target	+0.8%	Take profit execution limit
Stop Loss	-0.4%	2:1 Risk-to-Reward ratio
Time Restriction	1:30 PM IST	No new entries allowed after this time
Hard Square-Off	3:10 PM IST	All outstanding positions closed via market orders